Introducing...

Build It Green!

BUILD IT GREEN

An Interactive Storyline Unit
for Energy Literacy

A New Place-Based Energy Unit for Science Educators

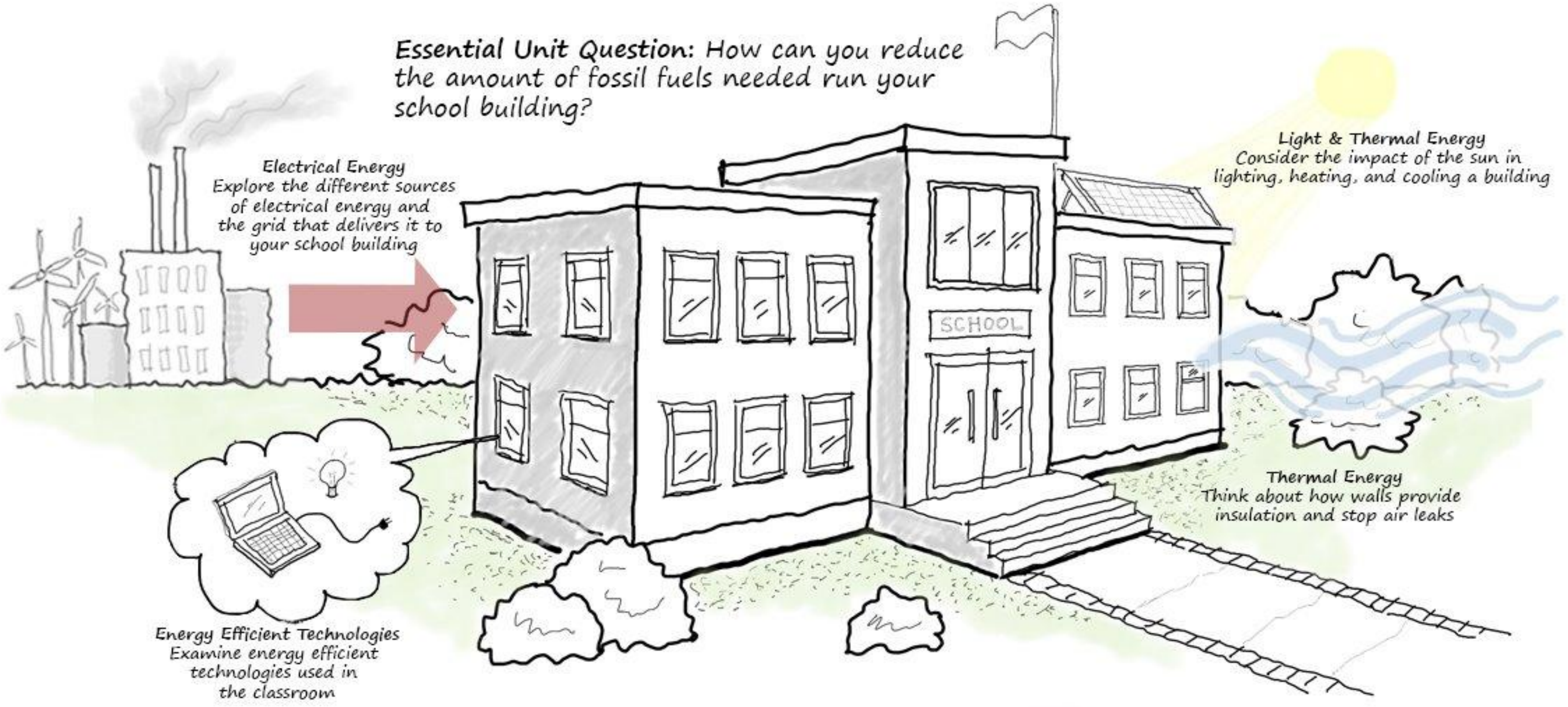
Jessica Justice, Yupei Duan, Caiden Webb, & Laura Zangori

This material is based upon work supported by the National Science Foundation (Award No 2201204). Any opinions, findings, and conclusions or recommendations expressed in this publication are those of the author(s) and do not necessarily reflect the views of the NSF

Teaching Energy Using the School Building

The school building is a place-based environment to learn about energy.

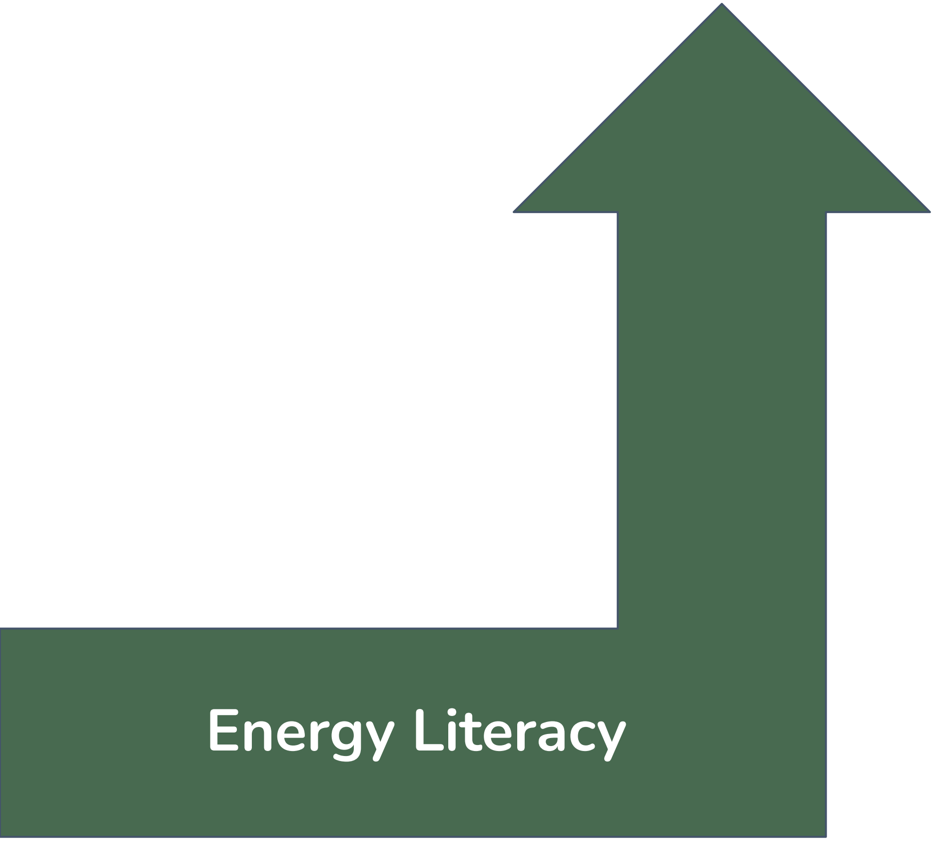
Scientific Modeling supports locating and tracing energy systems - key for energy literacy.



Phase 1: Energy in Your Environment (EYE)

Phase 1: Energy in Your Environment (EYE)

EYE Statistically Significant Learning Gains



After completing the unit:

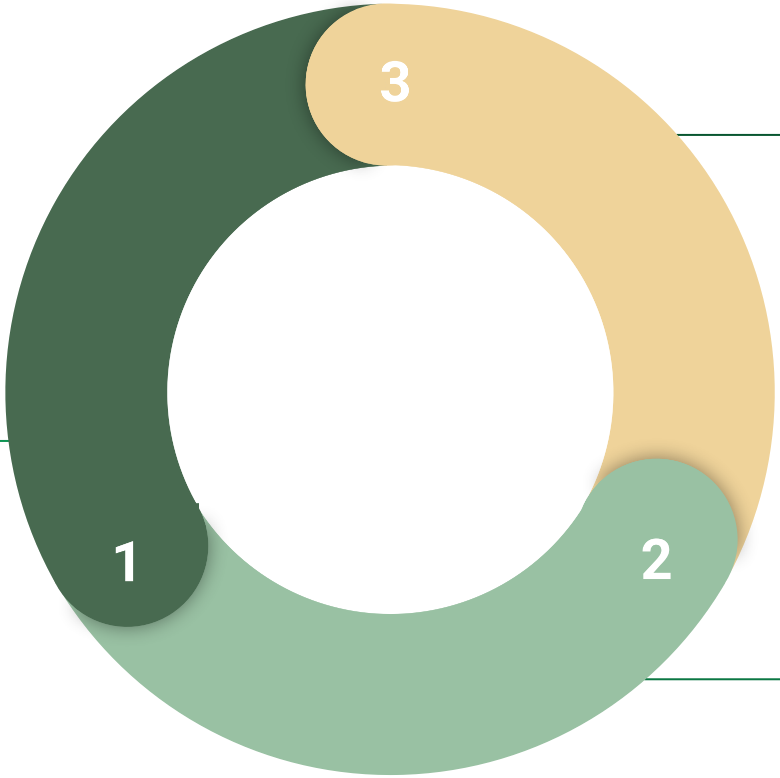
1. Students developed a more complex **understanding of energy flow**

2. Improved students' **ability to trace energy** throughout the energy system

The Evolution of BIG!

2020 - 2024: Development and Test of *Energy in Your Environment* (EYE)

- Findings show that students' energy learning improved!
- Student awareness of energy behaviors - both personal and in buildings - improved



2024 - 2026: BIG! Pilot Test

In this phase, we will support teachers as they pilot the unit. We will utilize their feedback to make revisions to improve the unit.

2022 - 2024: Development of *Build It Green!* (BIG!)

Phase 2: Developing BIG!

- Standards-aligned unit that teaches energy transfer (light, heat, and electrical) through engineering design.
  - Standard Alignment with NGSS and Missouri Learning Standards
- Hands-on investigations using your own school building
- Digital and hands-on simulations to understand energy flow and propose energy efficient building designs.

Our Goals

1. Supporting middle school students' understandings of energy transfer

2. Supporting middle school teachers in teaching place-based energy transfer

3. Creating real world, local situations that can be used to help students understand the importance of energy transfer in buildings

Unit Design

- Professionally designed with the help of BSCS Science Learning
- Place-based, storyline unit (Reiser, 2021) with anchoring phenomenon selected by students for students
- Modifiable to fit the needs of teachers and students everywhere

Lesson 1

Green Schools

How can we design a green school?

BUILD IT GREEN

An Interactive Storyline Unit for Energy Literacy

This material is based upon work supported by the National Science Foundation (Award No 2201204). Any opinions, findings, and conclusions or recommendations expressed in this publication are those of the author(s) and do not necessarily reflect the views of the NSF.

Schools & Energy

It is estimated that schools across the US spend over $6 BILLION annually on energy or about **$100 per student per year**.

- If a school district has 100 students, what is their energy bill?
- What if they have 1,000 students? Or, 10,000 students?

What do you do every day in our school that uses energy?

Schools & Energy

But the US Department of Energy estimate that 30% of this energy cost that schools have is actually “wasted” energy. **This means for every $100 spent per student, $30 is wasted and not used for learning.**

- What do you think “wasted” energy is? How could a school waste energy?
- What could a school do with that money they would save if they didn’t have wasted energy?

Why Green Schools?

Green schools:

- Use **less energy** than regular schools.
- Use **less water** than regular schools.
- Use **less toxic chemicals** in building materials.
- Help people **recycle and compost** their waste.
- **Save schools money** on heating, cooling, and other electricity costs.
- Have interesting building features like **lots of windows and open-air spaces** for students.
- Use **local or native plants** for landscaping, shade, and wildlife.

What is a green school?

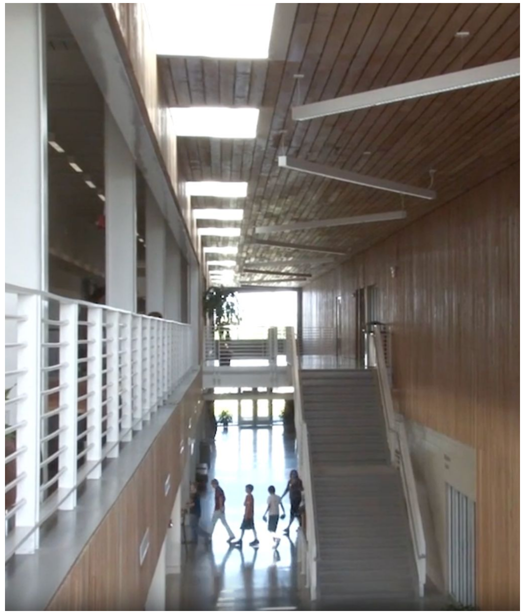
Take a look at the case study schools on the handouts.

How are they similar to your school?
How are they different?

Case Study 1: Kiowa County School, KS

School Information

- PreK - 12th
- 410 students
- LEED Platinum School
- On site wind turbine
- Built after 2007 tornado destroyed the school.
- Rural area



What are your thoughts?

Case Study 2: The Redding School of the Arts, CA

School Information

- K-8th
- 573 Students
- Net zero energy use
- First ever LEED platinum school
- 600 seat open-air theater
- Suburban area

What are your thoughts?

Case Study 3: Wydown Middle School, MO




School Information

- 6th - 8th
- Approximately 600 students
- LEED Gold Certified
- Solar panels
- Underground parking with green roof playing fields.
- Suburban area

What are your thoughts?

Sharing related things we've heard about

Have you ever heard or seen anything similar to these 'green' schools that reduce energy use or environmental impact?

- What is it?
- Where did you hear about it or see it?

Things similar to 'green' schools

Sharing how might you design a green school?

If you could design a green school, how would you do it?

- Take a few minutes to sketch your own design
- When you're done, chat with the person next to you about your design. You should consider the questions to the right as you discuss.

- **Do you have similar things?**
- **Do you have different things?**
- **Why do you think these designs may or may not work?**

Mapping Out the Unit

Setting the Stage

Introducing green schools and understanding the classroom as an energy system

Introducing the Design Challenge

Determining the type of space we will design, identifying criteria and constraints, creating an initial design

Understanding Energy

Learning about traditional and renewable energy sources, energy generation, and energy efficiency (HVAC, insulation, etc.)

Putting It All Together

Designing the final build, simulating the build and reviewing results, constructing a three-dimensional model

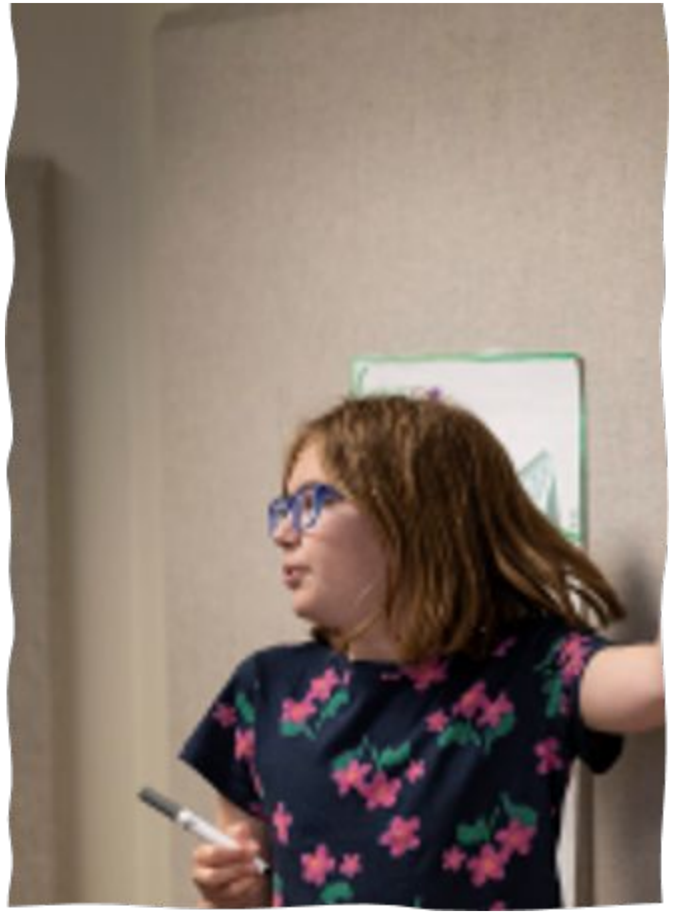
Phase 3: Piloting BIG!

$3 million — Awarded by the NSF to create, test, and publish BIG!

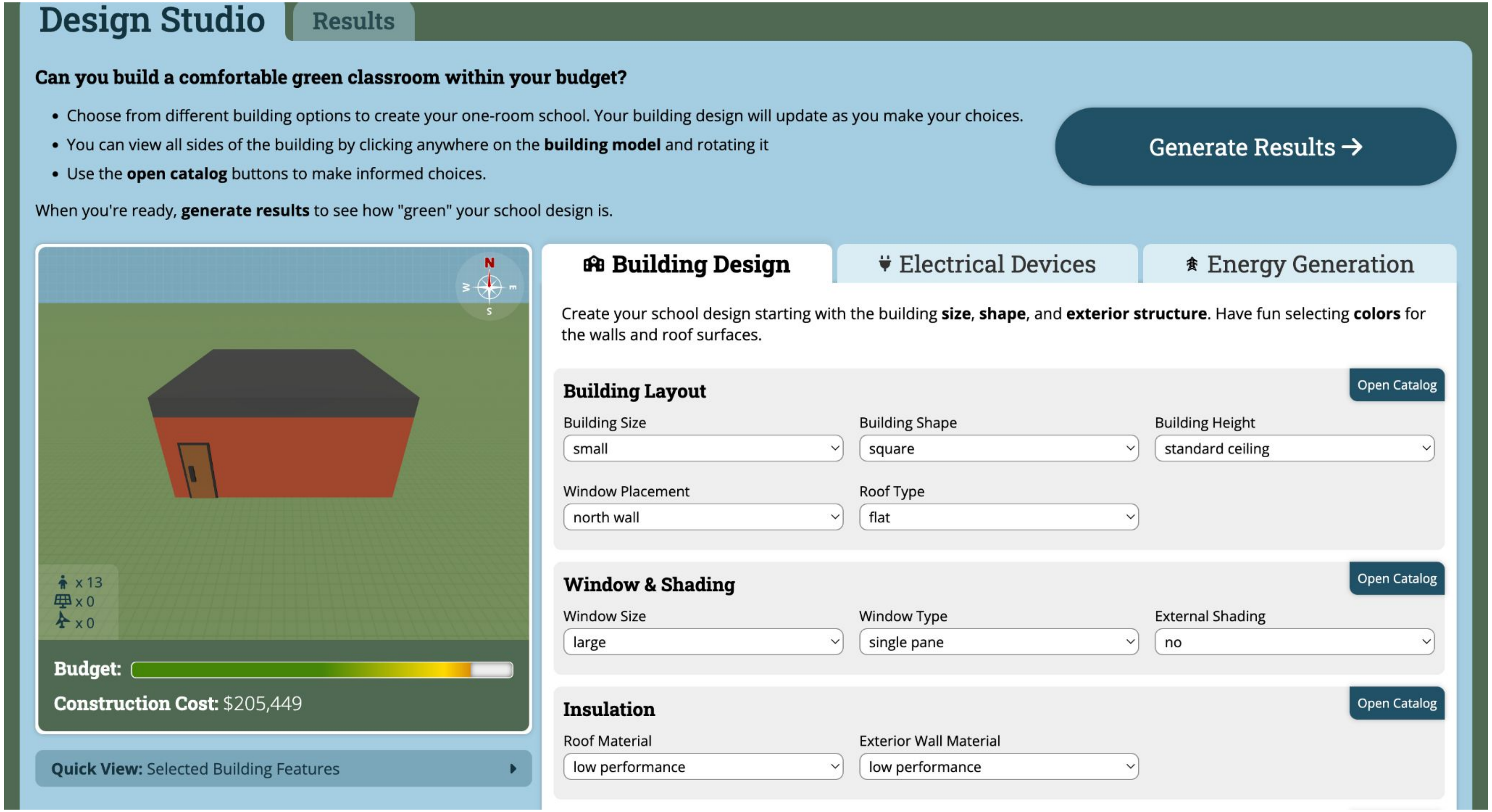
14 — Scholars, architects, content creators, and curriculum designers supporting the development and pilot testing of BIG!

28 — Teachers actively piloting BIG! this school year.

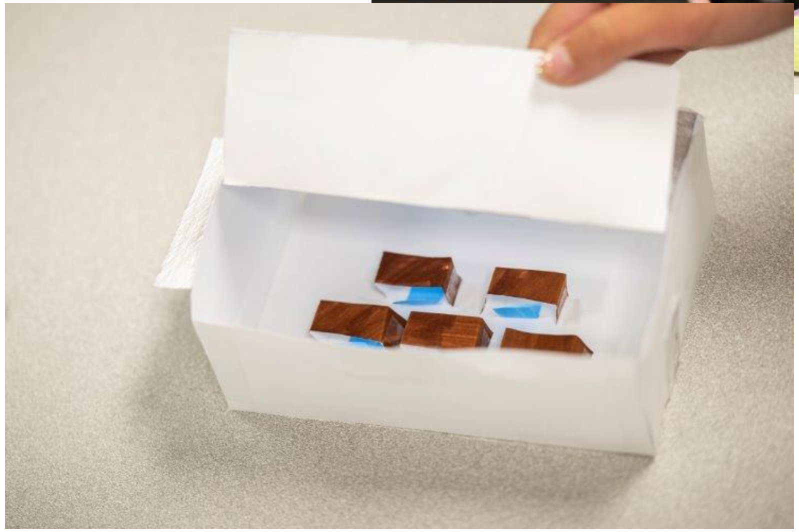
Piloting BIG! at Missouri S&T (July 2024)

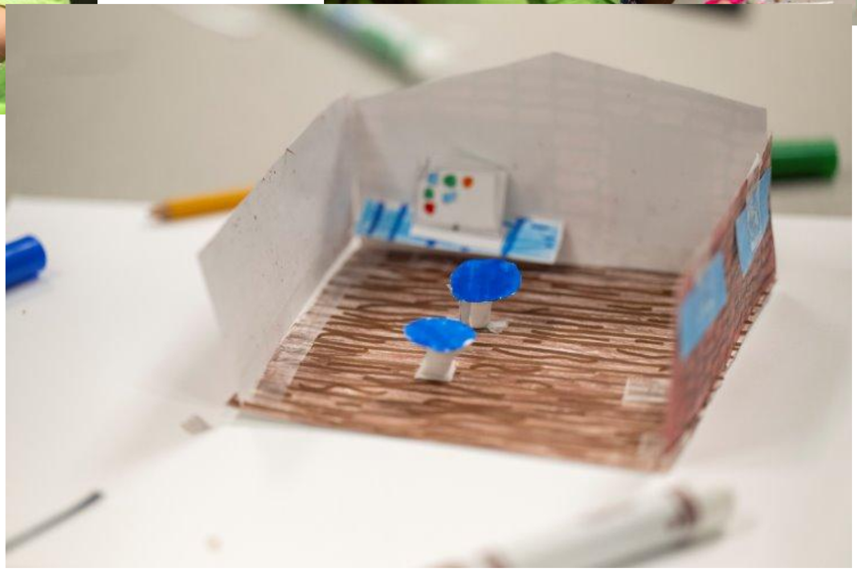


The Final Simulation



Building Sustainable Schools





Why bring BIG! to your school?

- BIG! is new, hands-on, and low-cost!
- BIG! transcends the traditional classroom! Your school is no longer a “venue” for learning - it’s a vital part of the learning experience!
- BIG! works to level the playing field, giving every student the same experience!
- BIG! introduces STEM careers and trades to your community’s biggest assets - its students!

Thank you!

We know that your students will love BIG! as much as we do!

Scan the QR code to learn more about partnering with us to bring BIG! to your school!